

CP303 Project Report

Mechanistic Kinetic modeling of Furfural Hydrodeoxygenation to Produce 2-Methylfuran in a Fixed Bed Reactor

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ABSTRACT

The catalytic conversion of furfural into 2-methylfuran has gained significant attention due to the potential application of 2-methylfuran as a renewable biofuel and value-added chemical derived from lignocellulosic biomass. In the present work, kinetic modelling of furfural hydrogenation to 2-methylfuran over CuFe_2O_4 and $\text{Cu/Fe}_2\text{O}_3$ catalysts was investigated using power law and Langmuir–Hinshelwood mechanistic approaches. The reaction pathway involved sequential hydrogenation of furfural to furfuryl alcohol followed by hydrodeoxygenation to 2-methylfuran. Single-site and dual-site Langmuir–Hinshelwood mechanistic kinetic models were developed based on adsorption, desorption, and surface reaction steps. The kinetic and thermodynamic parameters were estimated using nonlinear regression analysis of packed bed reactor experimental data collected over a temperature range of 190–250°C under atmospheric pressure. Modified Arrhenius and van't Hoff equations were employed for estimation of temperature-dependent kinetic and adsorption parameters. The estimated activation energies indicated that the deoxy hydrogenation step was more energy demanding than furfural hydrogenation. Negative adsorption enthalpy and entropy values confirmed exothermic adsorption and reduced molecular randomness on catalyst active sites. Statistical analyses including confidence intervals, t-tests, F-tests, RMSE analysis, and correlation matrix evaluation were performed to validate the developed models. Among the investigated models, the dual-site Langmuir–Hinshelwood mechanism demonstrated improved thermodynamic consistency, lower parameter correlation, and better statistical significance compared to the single-site mechanism. Heat and mass transfer limitation analyses confirmed the absence of transport limitations, indicating that the observed reaction rates represented intrinsic catalytic kinetics. The developed dual-site mechanistic model successfully described the catalytic conversion of furfural to 2-methylfuran and provided insight into adsorption and surface reaction phenomena governing the reaction pathway.

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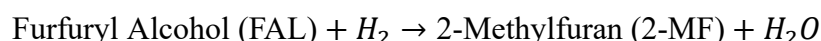
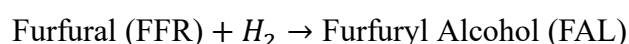
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1. INTRODUCTION

The increasing depletion of fossil fuel reserves and growing environmental concerns have accelerated the search for sustainable alternatives for fuels and chemicals. Biomass has emerged as a promising renewable carbon source due to its abundance, renewability, and lower environmental impact. Among biomass-derived compounds, furfural is considered an important platform molecule obtained from the dehydration of pentose sugars present in lignocellulosic biomass. Because of its reactive aldehyde and furan functional groups, furfural can be converted into several valuable chemicals and biofuels.

Among the products derived from furfural, 2-methylfuran (2-MF) has gained significant attention as a potential biofuel and fuel additive. 2-MF possesses desirable fuel properties such as high-octane number, high energy density, low water solubility, and good combustion characteristics. Compared to conventional biofuels like ethanol, 2-MF exhibits better fuel compatibility and improved energy efficiency. Consequently, the catalytic upgrading of furfural to 2-MF has become an important area of research in biomass valorisation.

The production of 2-methylfuran from furfural generally occurs through sequential hydrogenation reactions. In the first step, furfural is hydrogenated to furfuryl alcohol (FAL), which is further converted to 2-methylfuran through hydrodeoxygenation. The reaction pathway can be represented as:



The conversion and selectivity of these reactions strongly depend on catalyst properties, temperature, hydrogen pressure, and adsorption behaviour of reactants on catalyst surfaces. Copper-based catalysts are widely used for furfural hydrogenation because of their favourable selectivity toward furfuryl alcohol and 2-MF. In the present work, CuFe_2O_4 and $\text{Cu/Fe}_2\text{O}_3$ catalyst systems are considered for the catalytic conversion of furfural to 2-methylfuran. The interaction between copper species and iron oxide phases can improve catalytic activity, adsorption characteristics, and product selectivity.

Since furfural hydrogenation is a heterogeneous catalytic process, adsorption and surface reactions play an important role in determining overall reaction rates. Empirical kinetic models often fail to represent these surface interactions accurately. Therefore, mechanistic kinetic models based on adsorption-reaction mechanisms are preferred for describing such catalytic

systems. The Langmuir-Hinshelwood (L-H) approach is one of the most commonly used methods for modelling heterogeneous catalytic reactions. The model assumes that reactants adsorb on catalyst active sites and react through surface-mediated elementary steps. Such models provide physically meaningful parameters including adsorption constants, activation energies, and intrinsic reaction rate constants.

Several researchers have investigated furfural hydrogenation using copper-based and bimetallic catalysts. Sitthisa et al. reported that adsorption of furfural on copper surfaces significantly influences hydrogenation pathways and selectivity toward furfuryl alcohol and 2-methylfuran. Other studies involving bimetallic catalysts showed that oxygen vacancies and metal-support interactions strongly affect catalytic performance and product distribution. Mechanistic studies have also shown that Langmuir–Hinshelwood–Hougen–Watson (LHHW) models provide better agreement with experimental data compared to simple power-law models because they account for adsorption effects and competitive surface interactions.

Despite significant progress in catalyst development and experimental studies, detailed mechanistic kinetic investigations of 2-MF production remain limited. In particular, the effects of reversible and irreversible reaction pathways, adsorption behaviour, and denominator powers in Langmuir–Hinshelwood formulations require further analysis. Therefore, the development of reliable mechanistic kinetic models is necessary for understanding intrinsic catalytic behaviour and optimizing reactor performance.

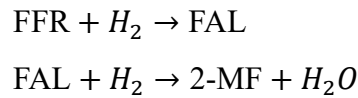
The present work focuses on the development of Langmuir–Hinshelwood mechanistic kinetic models for the production of 2-methylfuran from furfural. The study includes derivation of kinetic rate expressions, estimation of kinetic and adsorption parameters using experimental data, and evaluation of temperature effects on reaction kinetics. Different mechanistic assumptions involving reversible and irreversible reactions are compared to identify the most suitable model for predicting reactor performance and product selectivity.

2. DEVELOPMENT OF KINETIC MODELS

2.1 Power Law Kinetic Model

The power law kinetic model is one of the simplest approaches used for describing catalytic reaction systems. In this model, the reaction rate is expressed as a function of reactant partial pressures raised to empirical reaction orders. Although the model does not explicitly account for adsorption and surface interactions, it is useful for preliminary analysis of reaction kinetics

and estimation of apparent kinetic parameters. The catalytic conversion of furfural to 2-methylfuran occurs through two consecutive hydrogenation reactions:



In the present study, hydrogen was supplied in large excess compared to furfural concentration. Due to the excess hydrogen conditions, the variation in hydrogen partial pressure during the reaction was assumed to be negligible. Therefore, the hydrogen concentration term was incorporated into the apparent rate constant, resulting in pseudo-order kinetic expressions. Accordingly, the power law rate expressions for the two reactions were written as:

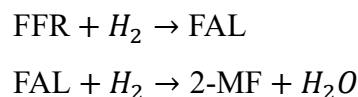
$$\begin{aligned}r_{\text{FFR}} &= k_c P_{\text{FFR}}^m \\ r_{\text{FAL}} &= k_c P_{\text{FFR}}^m - k_e P_{\text{FAL}}^n\end{aligned}$$

where r_{FFR} , P_{FFR} and r_{FAL} , P_{FAL} represent the rates and the partial pressures of furfuryl alcohol and 2-methylfuran formation, respectively. k_c and k_e are the apparent rate constants for the two reactions, while m and n are the apparent reaction orders.

The power law model provides a simplified representation of the reaction system and is useful for estimating apparent kinetic parameters from experimental data. However, since the model neglects adsorption effects and catalyst surface interactions, it may not accurately describe the intrinsic kinetics of heterogeneous catalytic hydrogenation reactions. Therefore, a more rigorous mechanistic model based on the Langmuir–Hinshelwood approach was further developed in the present work.

2.2 Langmuir–Hinshelwood Mechanistic Kinetic Model

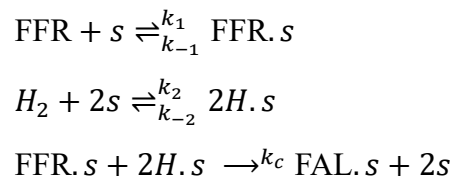
The catalytic conversion of furfural to 2-methylfuran is a heterogeneous catalytic process involving adsorption, surface reaction, and desorption steps over catalyst active sites. In the present work, Langmuir–Hinshelwood (L–H) mechanistic kinetic models were developed to describe the intrinsic kinetics of furfural hydrogenation over CuFe_2O_4 and $\text{Cu/Fe}_2\text{O}_3$ catalysts. The overall reaction pathway consists of two consecutive reactions involving hydrogenation of furfural to furfuryl alcohol followed by hydrodeoxygenation of furfuryl alcohol to 2-methylfuran. The overall reactions are represented as:



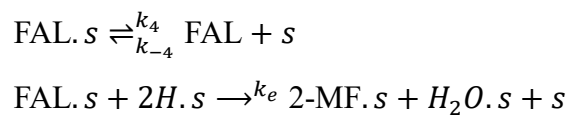
The overall heat of reaction was approximately 50 kJ mol⁻¹, indicating the exothermic nature of the hydrogenation process. The experiments were conducted under isothermal operating conditions over the investigated temperature range.

Two different mechanistic approaches were considered in the present study. The first mechanism was based on a single-site Langmuir–Hinshelwood model in which all reactants, intermediates, and products adsorb on the same type of catalyst active site. The second mechanism was based on a dual-site adsorption model involving two different active sites on the catalyst surface.

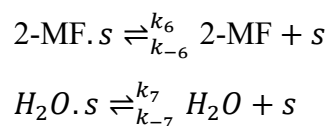
For the single-site Langmuir–Hinshelwood mechanism, it was assumed that all reactants, intermediates, and products adsorb on the same type of catalyst active site. The adsorption and desorption steps were considered to be quasi-equilibrated, while the surface reactions were assumed to be rate determining. Hydrogen adsorption on the catalyst surface was considered to occur dissociatively. External and internal mass transfer resistances were neglected, and the catalyst surface was assumed to be energetically uniform. The third reaction corresponding to furfural hydrogenation and the fifth reaction corresponding to deoxy hydrogenation were considered as the rate-determining steps. The reactor was assumed to operate under isothermal conditions with negligible pressure drop across the catalyst bed. The elementary reaction steps for the single-site Langmuir–Hinshelwood mechanism are proposed below:



(Furfural hydrogenation: Rate-determining step)



(Deoxy hydrogenation: Rate-determining step)



The adsorption equilibrium constants for the adsorbed species are represented as:

$$K_{FFR} = \frac{k_1}{k_{-1}} = \frac{C_{FFR.s}}{P_{FFR}C_s}$$

$$K_H = \frac{k_2}{k_{-2}} = \frac{(C_{H.s})^2}{P_{H_2}(C_s)^2}$$

$$K_{FAL} = \frac{k_4}{k_{-4}} = \frac{C_{FAL.s}}{P_{FAL}C_s}$$

$$K_{MF} = \frac{k_6}{k_{-6}} = \frac{C_{MF.s}}{P_{MF}C_s}$$

$$K_{H_2O} = \frac{k_7}{k_{-7}} = \frac{C_{H_2O.s}}{P_{H_2O}C_s}$$

From the above relations:

$$C_{FFR.s} = K_{FFR}P_{FFR}C_s$$

$$C_{H.s} = \sqrt{K_H P_{H_2} C_s}$$

$$C_{FAL.s} = K_{FAL}P_{FAL}C_s$$

$$C_{MF.s} = K_{MF}P_{MF}C_s$$

$$C_{H_2O.s} = K_{H_2O}P_{H_2O}C_s$$

The site balance equation for the single-site mechanism can be expressed as:

$$\text{For FFR: } C_t = C_s + C_{FFR.s} + C_{H.s} + C_{FAL.s}$$

$$\text{For FAL: } C_t = C_s + C_{H.s} + C_{FAL.s} + C_{MF.s} + C_{H_2O.s}$$

Substituting the adsorption relationships into the site balance equation respectively:

$$C_t = C_s \left[1 + K_{FFR}P_{FFR} + \sqrt{K_H P_{H_2}} + K_{FAL}P_{FAL} \right]$$

$$C_t = C_s \left[1 + \sqrt{K_H P_{H_2}} + K_{FAL}P_{FAL} + K_{MF}P_{MF} + K_{H_2O}P_{H_2O} \right]$$

Therefore, the concentration of vacant active sites becomes:

$$C_s = \frac{C_t}{1 + K_{FFR}P_{FFR} + \sqrt{K_H P_{H_2}} + K_{FAL}P_{FAL}}$$

$$C_s = \frac{C_t}{1 + \sqrt{K_H P_{H_2}} + K_{FAL}P_{FAL} + K_{MF}P_{MF} + K_{H_2O}P_{H_2O}}$$

Since the third and fifth reaction corresponding to furfural hydrogenation and deoxy hydrogenation is the rate-determining step, the rate expressions respectively can be written as:

$$r_{FFR} = k_3 C_{FFR.s} (C_{H.s})^2$$

$$r_{FAL} = k_e C_{FAL.s} (C_{H.s})^2$$

Substituting the adsorption relationships:

$$r_{FFR} = k_c (K_{FFR} P_{FFR} C_s) \left(\sqrt{K_H P_{H_2} C_s} \right)^2$$

$$r_{FFR} = k_c K_{FFR} K_H P_{FFR} P_{H_2} C_s^3$$

$$r_{FAL} = k_e (K_{FAL} P_{FAL} C_s) \left(\sqrt{K_H P_{H_2} C_s} \right)^2$$

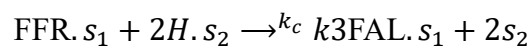
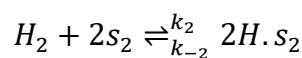
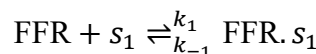
$$r_{FAL} = k_e K_{FAL} K_H P_{FAL} P_{H_2} C_s^3$$

Finally, substituting the expression for vacant site concentration:

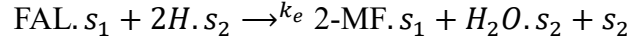
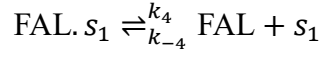
$$r_{FFR} = \frac{k_c K_{FFR} K_H P_{FFR} P_{H_2}}{(1 + K_{FFR} P_{FFR} + \sqrt{K_H P_{H_2}} + K_{FAL} P_{FAL})^3}$$

$$r_{FAL} = \frac{k_e K_{FAL} K_H P_{FAL} P_{H_2}}{(1 + K_{FAL} P_{FAL} + \sqrt{K_H P_{H_2}} + K_{MF} P_{MF} + K_{H_2O} P_{H_2O})^3}$$

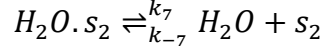
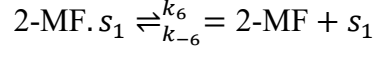
For the dual-site Langmuir–Hinshelwood mechanism, it was assumed that adsorption and desorption steps are quasi-equilibrated while the surface reactions are rate determining. Hydrogen adsorption occurs dissociatively on hydrogen-active sites, whereas furfural and other oxygenated species adsorb on separate organic adsorption sites. External and internal mass transfer limitations were neglected, and the catalyst surface was assumed to be energetically uniform. The third reaction corresponding to furfural hydrogenation and the fifth reaction corresponding to deoxy hydrogenation were considered as the rate-determining steps. The reactor was assumed to operate under isothermal conditions with negligible pressure drop. The elementary reaction steps for the dual-site mechanism are represented below:



(Furfural hydrogenation: Rate-determining step)



(Deoxy hydrogenation: Rate-determining step)



The adsorption equilibrium constants are represented as:

$$K_{FFR} = \frac{k_1}{k_{-1}} = \frac{C_{FFR.s_1}}{P_{FFR}C_{s_1}}$$

$$K_H = \frac{k_2}{k_{-2}} = \frac{(C_{H.s_2})^2}{P_{H_2}(C_{s_2})^2}$$

$$K_{FAL} = \frac{k_4}{k_{-4}} = \frac{C_{FAL.s_1}}{P_{FAL}C_{s_1}}$$

$$K_{MF} = \frac{k_6}{k_{-6}} = \frac{C_{MF.s_1}}{P_{MF}C_{s_1}}$$

$$K_{H_2O} = \frac{k_7}{k_{-7}} = \frac{C_{H_2O.s_2}}{P_{H_2O}C_{s_2}}$$

From the above relations:

$$C_{FFR.s_1} = K_{FFR}P_{FFR}C_{s_1}$$

$$C_{H.s_2} = \sqrt{K_H P_{H_2} C_{s_2}}$$

$$C_{FAL.s_1} = K_{FAL}P_{FAL}C_{s_1}$$

$$C_{MF.s_1} = K_{MF}P_{MF}C_{s_1}$$

$$C_{H_2O.s_2} = K_{H_2O}P_{H_2O}C_{s_2}$$

The site balance for organic adsorption sites is:

$$\text{For FFR: } C_{t1} = C_{s_1} + C_{FFR.s_1} + C_{FAL.s_1}$$

$$\text{For FAL: } C_{t1} = C_{s_1} + C_{FAL.s_1} + C_{MF.s_1}$$

Substituting adsorption relationships respectively:

$$C_{t1} = C_{s_1} (1 + K_{FFR}P_{FFR} + K_{FAL}P_{FAL})$$

$$C_{t1} = C_{s1}(1 + K_{MF}P_{MF} + K_{FAL}P_{FAL})$$

Therefore:

$$C_{s1} = \frac{C_{t1}}{1 + K_{FFR}P_{FFR} + K_{FAL}P_{FAL}}$$

$$C_{s1} = \frac{C_{t1}}{1 + K_{FAL}P_{FAL} + K_{MF}P_{MF}}$$

Similarly, the site balance for hydrogen adsorption sites becomes:

$$\text{For FFR: } C_{t2} = C_{s2} + C_{H.s2}$$

$$\text{For FAL: } C_{t2} = C_{s2} + C_{H.s2} + C_{H_2O.s2}$$

Substituting hydrogen adsorption relationship respectively:

$$C_{t2} = C_{s2}(1 + \sqrt{K_H P_{H_2}})$$

$$C_{t2} = C_{s2}(1 + \sqrt{K_H P_{H_2}} + K_{H_2O} P_{H_2O})$$

Therefore:

$$C_{s2} = \frac{C_{t2}}{1 + \sqrt{K_H P_{H_2}}}$$

$$C_{s2} = \frac{C_{t2}}{1 + \sqrt{K_H P_{H_2}} + K_{H_2O} P_{H_2O}}$$

Since the third and fifth reactions are rate determining, the rate expression for furfural hydrogenation and deoxy hydrogenation and becomes:

$$r_{FFR} = k_c C_{FFR.s1} (C_{H.s2})^2$$

$$r_{FAL} = k_e C_{FAL.s1} (C_{H.s2})^2$$

Substituting adsorption relationships:

$$r_{FFR} = k_c K_{FFR} K_H P_{FFR} P_{H_2} C_{s1} C_{s2}^2$$

$$r_{FAL} = k_e K_{FAL} K_H P_{FAL} P_{H_2} C_{s1} C_{s2}^2$$

Finally substituting the vacant site concentrations:

$$r_{FFR} = \frac{k_c K_{FFR} K_H P_{FFR} P_{H_2}}{(1 + K_{FFR} P_{FFR} + K_{FAL} P_{FAL})(1 + \sqrt{K_H P_{H_2}})^2}$$

$$r_{FAL} = \frac{k_e K_{FAL} K_H P_{FAL} P_{H_2}}{(1 + K_{FAL} P_{FAL} + K_{MF} P_{MF})(1 + \sqrt{K_H P_{H_2}} + K_{H_2O} P_{H_2O})^2}$$

The kinetic models LH-I and LH-II both contain 7 parameters (2 rate constants and 5 equilibrium constants). The kinetic model developed by Langmuir-Hinshelwood method generally contains a large number of adjustable parameters. In addition, strong correlation exists among many of the parameters, for example, heats of adsorption and entropies of adsorption. Therefore, the values of heats of adsorption were fixed with the values reported in literature.

3. ESTIMATION OF KINETIC MODEL PARAMETERS

Experimental data for the hydrogenation of furfural to 2-methylfuran were obtained from packed bed reactor experiments carried out over CuFe_2O_4 and $\text{Cu/Fe}_2\text{O}_3$ catalysts under different operating conditions. The experiments were performed at temperatures of 190°C, 210°C, 230°C, and 250°C under atmospheric pressure. The raw experimental data consisted of hydrogen flow rate, nitrogen flow rate, H_2/FFR ratio, furfural feed flow rate, catalyst amount, temperature, WHSV, furfural conversion, and yields of furfuryl alcohol and 2-methylfuran. These raw experimental measurements were processed to calculate molar flow rates, partial pressures, and reaction rates required for kinetic model development and parameter estimation.

The calculated quantities obtained from the raw data included inlet and outlet molar flow rates of hydrogen, nitrogen, furfural, furfuryl alcohol, 2-methylfuran, and water. The processed variables included $F_{H_2,in}$, $F_{H_2,out}$, F_{N_2} , F_{FFR} , F_{FAL} , F_{MF} , F_T , $F_{FFR,out}$, F_{H_2O} , P_{FFR} , P_{FAL} , P_{2-MF} , P_{H_2O} , P_{H_2} , and P_{N_2} . These calculated quantities were subsequently used in the Langmuir–Hinshelwood kinetic rate expressions.

The outlet molar flow rates of products and reactants were calculated using furfural conversion and product yields obtained from experimental analysis. The total molar flow rate was determined by summing the outlet flow rates of all species present in the reactor stream.

Table 3.1 Flow Rate Calculations

Parameter	Formula	Example Calculation
$F_{FFR,out}$	$F_{FFR,out} = F_{FFR,in}(1 - X_{FFR})$	$0.00543(1 - 0.9518) = 0.000261 \text{ mol h}^{-1}$
F_{MF}	$F_{MF} = F_{FFR,in} \times Y_{MF}$	$0.00543 \times 0.7234 = 0.00393 \text{ mol h}^{-1}$
F_{FAL}	$F_{FAL} = F_{FFR,in} \times Y_{FAL}$	$0.005432 \times 0.1923 = 0.001045 \text{ mol h}^{-1}$
F_{H_2O}	$F_{H_2O} = F_{MF}$	$F_{H_2O} = 0.00393 \text{ mol h}^{-1}$
$F_{H_2,out}$	$F_{H_2,out} = F_{H_2,in} - (F_{FFR,in} \times X_{FFR} + F_{MF})$	$0.07411 - (0.005433 \times 0.9518 + 0.00393)$ $= 0.065202 \text{ mol h}^{-1}$
F_T	$F_T = \sum F_i$	$0.154668798 \text{ mol h}^{-1}$

The partial pressure of each species was calculated using the mole fraction relation:

$$P_i = \frac{F_i}{F_T} P_T$$

Table 3.2 Partial Pressure Calculations

Parameter	Formula	Example Calculation
P_{FFR}	$P_{FFR} = \frac{F_{FFR,out}}{F_T} P_T$	$\frac{0.000261869}{0.154668798} \times 1 = 0.001693097 \text{ bar}$
P_{FAL}	$P_{FAL} = \frac{F_{FAL}}{F_T} P_T$	$\frac{0.001044761}{0.154668798} \times 1 = 0.006754825 \text{ bar}$
P_{2-MF}	$P_{2-MF} = \frac{F_{MF}}{F_T} P_T$	$\frac{0.003930212}{0.154668798} \times 1 = 0.025410505 \text{ bar}$
P_{H_2O}	$P_{H_2O} = \frac{F_{H_2O}}{F_T} P_T$	$\frac{0.003930212}{0.154668798} \times 1 = 0.025410505 \text{ bar}$
P_{H_2}	$P_{H_2} = \frac{F_{H_2,out}}{F_T} P_T$	$\frac{0.065201958}{0.154668798} \times 1 = 0.421558572 \text{ bar}$
P_{N_2}	$P_{N_2} = \frac{F_{N_2}}{F_T} P_T$	$\frac{0.080299786}{0.154668798} \times 1 = 0.519172496 \text{ bar}$

The reaction rates were calculated on the basis of catalyst weight using the packed bed reactor design equation.

Table 3.3 Rate Calculations

Parameter	Formula	Example Calculation
r_{FFR}	$r_{FFR} = \frac{F_{FFR,in} - F_{FFR,out}}{W}$	$\frac{0.005432973 - 0.000261869}{1} = 0.005171103 \text{ mol h}^{-1} \text{g}_{cat}^{-1}$
r_{MF}	$r_{MF} = \frac{F_{MF}}{W}$	$\frac{0.003930212}{1} = 0.003930212 \text{ mol h}^{-1} \text{g}_{cat}^{-1}$
r_{FAL}	$r_{FAL} = \frac{F_{FAL}}{W}$	$\frac{0.001240891}{1} = 0.001240891 \text{ mol h}^{-1} \text{g}_{cat}^{-1}$

The processed experimental data consisting of molar flow rates, partial pressures, and reaction rates were subsequently used for kinetic model development and parameter estimation.

The temperature dependence of the kinetic rate constants was represented using a modified Arrhenius equation:

$$k = k_{ref} \exp \left[-\frac{E}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right]$$

where k_{ref} is the rate constant at the reference temperature T_{ref} , E is the activation energy, R is the universal gas constant, and T is the reaction temperature.

A reference temperature approach was adopted instead of the conventional Arrhenius equation to minimize correlation between the pre-exponential factor and activation energy during nonlinear regression. The reference temperature was selected as 503.15 K (230°C), corresponding approximately to the center of the experimental temperature range. Using a reference temperature improves numerical stability and enhances the reliability of estimated kinetic parameters.

The temperature dependence of adsorption equilibrium constants was represented using the van't Hoff type expression:

$$K_i = \exp \left[-\frac{\Delta H_i}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) + \frac{\Delta S_i}{R} \right]$$

where $K_{i,ref}$ is the adsorption equilibrium constant at the reference temperature, ΔH_i is the heat of adsorption, and ΔS_i is the entropy change associated with adsorption of species i .

The kinetic and adsorption parameters were estimated using nonlinear regression analysis by minimizing the difference between experimental and model-predicted reaction rates. MATLAB

optimization routines were employed for parameter estimation. The objective function used for regression analysis was the sum of squared errors (SSE):

$$SSE = \sum_{i=1}^N (y_{exp,i} - y_{model,i})^2$$

where $y_{exp,i}$ and $y_{model,i}$ represent experimental and predicted values, respectively.

The entropy parameters (ΔS_i) were directly estimated during nonlinear regression together with adsorption enthalpies and kinetic parameters. Rate constants obtained at different temperatures were further used to generate Arrhenius plots for validation and estimation of activation energies.

Confidence intervals for the estimated parameters were obtained from the covariance matrix generated during nonlinear regression. The 95% confidence interval for each parameter was calculated using parameter variance and residual error distribution. Smaller confidence intervals indicated better reliability and precision of the estimated kinetic parameters.

The goodness of fit between experimental and model-predicted values was evaluated using the coefficient of determination (R^2):

$$R^2 = 1 - \frac{\sum (y_{exp} - y_{model})^2}{\sum (y_{exp} - \bar{y}_{exp})^2}$$

Higher R^2 values indicated better agreement between model predictions and experimental observations.

Thermodynamic consistency of the adsorption parameters was also verified during regression analysis. Constraints were imposed on entropy changes (ΔS_i) to ensure physically meaningful adsorption behavior. Since adsorption generally reduces molecular randomness on the catalyst surface, entropy changes were restricted within physically realistic ranges. Adsorption enthalpies were also constrained to maintain exothermic adsorption characteristics.

Correlation matrices were analysed to evaluate parameter dependency and identifiability. High correlation between parameters indicates strong interdependence and reduced reliability of independent parameter estimation. Lower off-diagonal values in the correlation matrix indicated improved robustness and better parameter estimation quality for the developed kinetic model.

The estimated kinetic parameters, adsorption constants, activation energies, and thermodynamic properties obtained through regression analysis were subsequently used to validate the developed Langmuir–Hinshelwood kinetic models against experimental reactor data.

4. RESULTS AND DISCUSSION

The kinetic and thermodynamic parameters estimated using nonlinear regression analysis for the single-site Langmuir–Hinshelwood (LH1), dual-site Langmuir–Hinshelwood (LH2), and power law models are presented in Tables 4.1 and 4.2. The estimated parameters were evaluated using confidence intervals, RMSE values, t-tests, F-tests, and correlation matrix analysis to determine the statistical reliability and physical significance of the developed kinetic models.

4.1 Estimated Kinetic Parameters

The estimated kinetic parameters obtained from nonlinear regression are summarized in Table 4.1. The activation energies obtained for both Langmuir–Hinshelwood mechanisms and power law models were positive, confirming that the hydrogenation reactions are kinetically controlled and thermally activated processes.

The activation energy corresponding to the deoxy hydrogenation step (k_e) was consistently higher than that of furfural hydrogenation (k_c), indicating that conversion of furfuryl alcohol to 2-methylfuran is the more energy-intensive step in the overall reaction pathway. For the LH2 model, the activation energies for furfural hydrogenation and deoxy hydrogenation were estimated as 37.73 and 60.13 KJ/mol, respectively. Similarly, the power law model for furfuryl alcohol conversion produced a relatively high activation energy of 73.26 KJ/mol, further supporting the greater energy requirement for the second hydrogenation step.

The RMSE values obtained for all models were relatively low, indicating satisfactory agreement between experimental and model-predicted reaction rates. The LH-I model produced lower RMSE values for furfuryl alcohol prediction, whereas the power law model for furfural conversion exhibited the lowest RMSE among all evaluated models. However, statistical reliability and thermodynamic consistency must also be considered while selecting the most appropriate mechanistic model.

Table 4.1 Estimated Kinetic Parameters and F-Test Analysis for Different Kinetic Models

Reaction Mechanism and Model	Rate Constant	k _{ref}	(-95%) CI	(+95%) CI	E _i (KJ mol)	-95% CI	+95% CI	RMSE	F-Test Statistics
LH-I	k _c	0.12215	0.04097	0.20333	34.31	15.42	53.20	0.003035	(F=123.09), (p=0.0000)
Single Site	k _e	71.02	26.39	115.65	37.02	12.84	61.20	0.000542	(F=123.09), (p=0.0000)
LH-II	k _c	0.02253	0.02035	0.02471	37.73	29.88	45.58	0.003756	(F=74.08), (p=0.0000)
Dual-Site	k _e	123.72	74.12	173.32	60.13	29.80	90.46	0.000583	(F=74.08), (p=0.0000)
Power Law FFR (m=0.364)	k _c	0.087094	0.08328	0.09091	42.80	39.08	46.52	0.002304	—
Power Law FAL (n=0.533)	k _e	0.135041	0.09722	0.17286	73.26	38.11	108.41	0.001437	—

The F-test results indicated that both Langmuir–Hinshelwood models were statistically significant as overall regression models. The extremely small p-values confirmed that the developed kinetic models successfully captured the experimental trends observed during furfural hydrogenation.

4.2 Estimated Thermodynamic Parameters

The estimated thermodynamic parameters for adsorption equilibrium constants are presented in Table 4.2. The adsorption enthalpy values obtained for all adsorbed species were negative, indicating exothermic adsorption behaviour on the catalyst surface. Similarly, most entropy changes were negative, confirming reduced molecular randomness during adsorption on catalyst active sites.

The dual-site Langmuir–Hinshelwood model produced significantly lower confidence intervals and improved statistical significance compared to the single-site model. In contrast, the single-site model exhibited extremely large confidence intervals for several entropy parameters,

indicating poor parameter identifiability and strong parameter dependency during nonlinear regression.

Table 4.2 Estimated Thermodynamic Parameters and t-Test Analysis for Different Models

LH-I					LH-II				
Equilibrium Constant	ΔS_i (J/mol K)	(-95%) CI	(+95%) CI	t-Statistic / p-Value	ΔS_i (J/mol K)	(-95%) CI	(+95%) CI	t-Statistic / p-Value	ΔH_i (KJ/mol)
K_FFR	-96.03	NR	NR	(0.0552/0.9561)	-70.28	-79.22	-61.34	(15.48/ 0.0000)	-51.5
K_H	-133.75	NR	NR	(0.0370/ 0.9706)	-30.51	-36.35	-24.67	(10.254/ 0.0000)	-50
K_FAL	-31.81	NR	NR	(0.0153/0.9878)	-32.53	-37.68	-27.38	(-12.852/ 0.0000)	-28.9
K_MF	-84.27	NR	NR	(4.8892E-8/1.0000)	-17.57	NR	NR	(-0.2821/ 0.7783)	-15
K_H2O	-84.80	NR	NR	(-0.0480/0.9618)	-38.87	-80.71	2.97	(-1.5349/ 0.1274)	-52

The t-test analysis showed that the entropy parameters estimated using the LH-II model were statistically more reliable than those obtained from the LH-I model. Parameters (dS1), (dS2), and (dS4) in the dual-site mechanism exhibited high t-statistics with near-zero p-values, indicating strong statistical significance at the 95% confidence level. However, several parameters in the LH1 model exhibited extremely low t-statistics and large p-values, confirming poor sensitivity of the model toward adsorption parameter estimation.

4.3 Correlation Matrix Analysis

Correlation matrix analysis was performed to evaluate parameter dependency and identifiability during nonlinear regression. The correlation matrix for the dual-site model showed moderate correlation among most thermodynamic parameters, whereas the single-site model exhibited near-perfect positive and negative correlation between several entropy parameters.

In the LH-I model, parameters (dS1), (dS2), and (dS4) were almost perfectly correlated, while (dS7) exhibited perfect negative correlation with these parameters. Such high parameter dependency indicates instability in nonlinear optimization and poor reliability of independent parameter estimation.

In contrast, the LH-II model showed comparatively lower parameter correlation, indicating improved robustness and better parameter identifiability. The use of the reference temperature form of the Arrhenius equation also reduced correlation between activation energies and reference rate constants.

Table 4.3 Correlation Matrix for Thermodynamic Parameters of Dual-Site (LH-II) Model

	dS1	dS2	dS4	dS6	dS7
dS1	1.000	-0.518	0.036	-0.032	0.067
dS2	-0.518	1.000	-0.169	-0.006	-0.051
dS4	0.036	-0.169	1.000	0.664	-0.756
dS6	-0.032	-0.006	0.664	1.000	-0.977
dS7	0.067	-0.051	-0.756	-0.977	1.000

Table 4.4 Correlation Matrix for Thermodynamic Parameters of Single-Site (LH1) Model

	dS1	dS2	dS4	dS6	dS7
dS1	1.000	1.000	1.000	-0.042	-1.000
dS2	1.000	1.000	1.000	-0.042	-1.000
dS4	1.000	1.000	1.000	-0.042	-1.000
dS6	-0.042	-0.042	-0.042	1.000	0.042
dS7	-1.000	-1.000	-1.000	0.042	1.000

The results clearly indicate that the dual-site model provides superior parameter identifiability and improved statistical robustness compared to the single-site mechanism.

4.4 Heat and Mass Transfer Limitation Analysis

Heat and mass transfer limitation analysis was performed to verify that the experimental data were obtained under kinetically controlled conditions. The calculated values of the Prater criterion, Weisz–Prater criterion, Mears criterion, and external heat transfer criterion were all significantly below the accepted limiting values.

Table 4.5 Heat and Mass Transfer Limitation Analysis

Criterion	Formula	Calc. Value	Limit	Interpretation
Internal Heat Transfer (Prater) $\Delta T_{max,particle}$	$\frac{D_{eff}(C_{A_S} - C_{A_C}) \Delta H_R}{\lambda_{eff}}$	0.00026436	< 1K	No limitation
External Heat Transfer (Film) $\Delta T_{max,film}$	$\frac{L(-r_{A_{obs}}) \Delta H_i}{h}$	0.02576	< 1K	No limitation
Internal Mass Transfer (Weisz–Prater) C_{WP}	$\frac{(-r_{A_{obs}})\rho_b R_C^2}{D_{eff}C_{A_S}}$	0.0059481	< 0.1	No limitation
External Mass Transfer (Mears)	$\frac{(-r_{A_{obs}})\rho_b R_C^n}{k_C C_A}$	0.0031723	< 0.15	No limitation

The negligible transport resistances indicate that the observed reaction rates represent intrinsic catalytic kinetics rather than diffusion-controlled behaviour. Therefore, the developed kinetic models can be considered suitable for describing the intrinsic reaction mechanism for conversion of furfural to 2-methylfuran.

4.5 Comparison with Literature

The kinetic parameters obtained in the present study were compared with previously reported studies on furfural hydrogenation and hydrodeoxygenation over copper-based catalysts. The activation energies obtained for both furfural hydrogenation and deoxy hydrogenation reactions were found to be within the range reported in the literature for biomass-derived furanic compound hydrogenation systems.

The activation energy corresponding to the furfural hydrogenation step (k_c) in the dual-site Langmuir–Hinshelwood model was estimated as 37.73 kJ/mol, while the activation energy for the deoxy hydrogenation step (k_e) was 60.13 kJ/mol. These values are comparable to the activation energies reported for Cu-based and bimetallic catalysts in previous studies.

Literature reports indicate that hydrogenation of the carbonyl group in furfural generally requires lower activation energy compared to the subsequent hydrodeoxygenation step leading to 2-methylfuran formation. Similar observations were reported by Sitthisa et al., where furfuryl alcohol was identified as the primary intermediate during furfural conversion over Cu-containing catalysts.

The higher activation energy obtained for deoxy hydrogenation in the present work suggests that removal of oxygen from furfuryl alcohol is the kinetically more demanding step. This observation agrees with previous mechanistic studies which reported that hydrogenolysis and hydrodeoxygenation reactions require stronger adsorption interactions and higher surface activation energy compared to simple carbonyl hydrogenation.

The adsorption enthalpy values estimated in the present study were negative for all adsorbed species, confirming exothermic adsorption behaviour. Similar trends have been reported in earlier Langmuir–Hinshelwood studies involving furfural hydrogenation over copper and supported metal catalysts. The negative entropy changes (ΔS_i) obtained for most adsorbed species further support the physical validity of the developed adsorption model. Negative entropy values indicate reduction in molecular randomness during adsorption, which is expected when gas-phase molecules become adsorbed and immobilized on catalyst active sites.

The entropy changes estimated using the dual-site model were found to be within physically realistic ranges and exhibited comparatively smaller confidence intervals than those obtained for the single-site model. In contrast, the single-site model produced extremely large confidence intervals and statistically insignificant entropy parameters, indicating poor parameter identifiability and strong parameter correlation. The improved statistical behaviour of the dual-site model suggests that adsorption of furfural, hydrogen, and intermediates occurs on different types of active sites rather than a single uniform surface.

Several literature studies have similarly reported that adsorption of hydrogen and oxygenated intermediates proceeds through separate adsorption centres over copper-based catalysts. The improved thermodynamic consistency and lower parameter correlation observed for the dual-site mechanism therefore agree well with the mechanistic interpretation proposed in previous studies.

Compared to empirical power law models, the Langmuir–Hinshelwood models developed in this work provide better mechanistic interpretation of adsorption and surface reaction

phenomena. The incorporation of adsorption equilibrium constants and thermodynamic constraints enabled physically meaningful estimation of intrinsic kinetic parameters.

The results obtained in the present study are therefore in good agreement with previously reported mechanistic and kinetic studies on furfural hydrogenation systems, while also providing additional insight into the role of dual active sites and adsorption thermodynamics governing the selective formation of 2-methylfuran.

4.6 Arrhenius and Parity Plot Analysis

The Arrhenius plots obtained for the kinetic rate constants of both LH1 and LH2 models showed approximately linear dependence between $\ln(k)$ and $(1/T - 1/T_{ref})$, confirming the validity of the modified Arrhenius relationship used during parameter estimation. The slopes of the plots corresponded to activation energies, while the intercepts represented reference rate constants at the selected reference temperature.

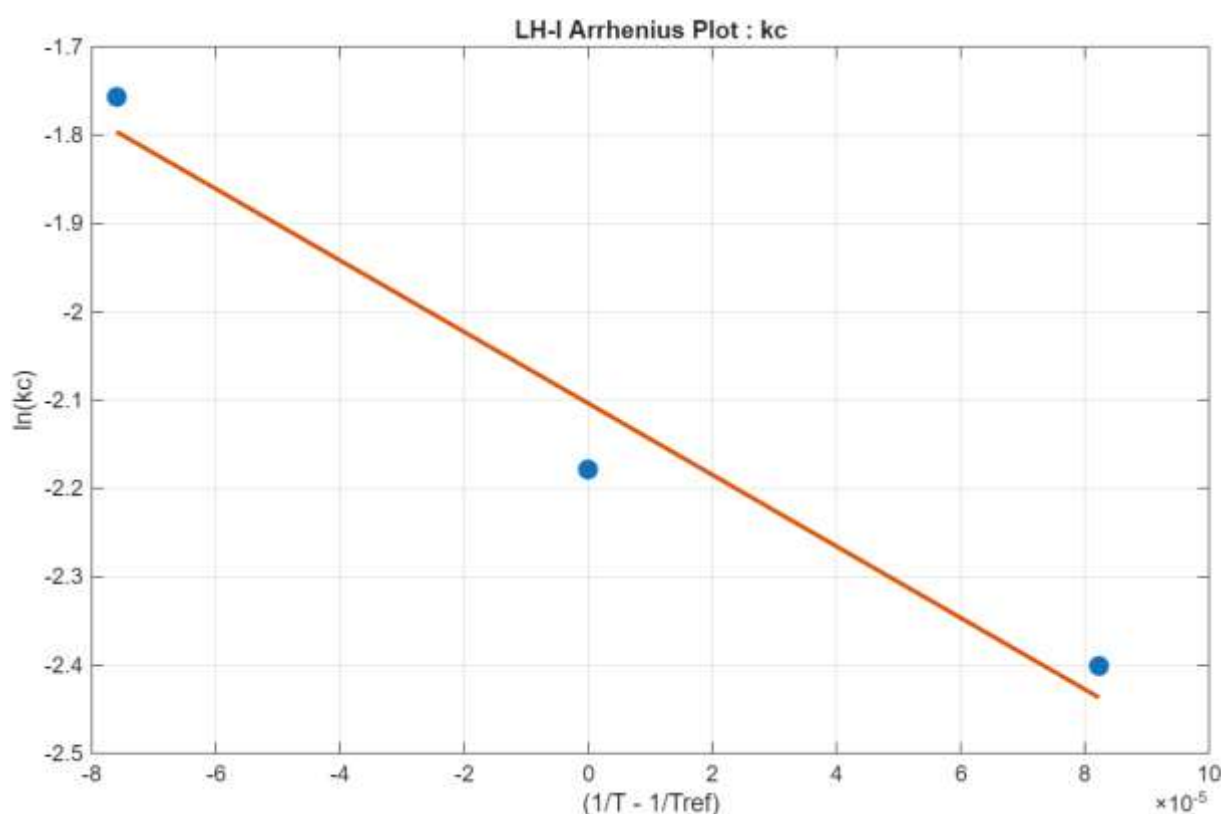


Figure 4.1 Arrhenius plot for furfural hydrogenation rate constant (k_c) using the single-site Langmuir–Hinshelwood (LH-I) model.

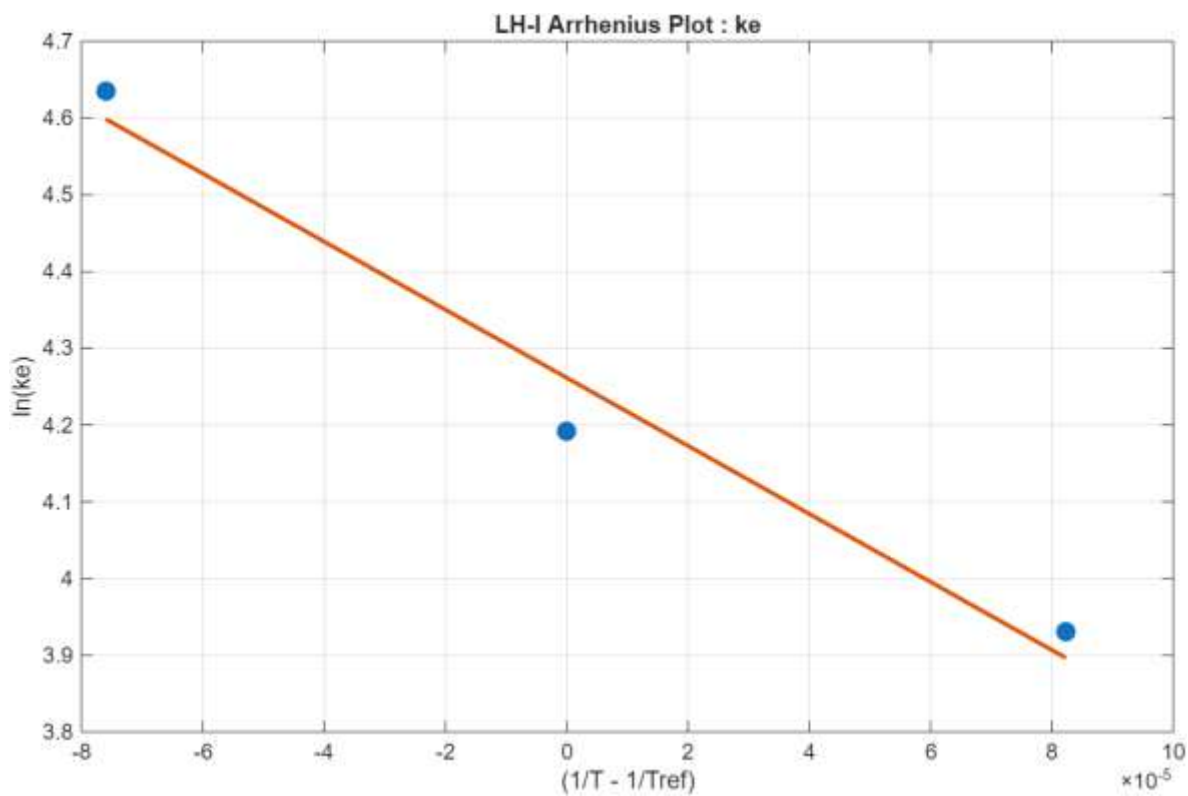


Figure 4.2 Arrhenius plot for deoxy hydrogenation rate constant (k_e) using the single-site Langmuir–Hinshelwood (LH-I) model.

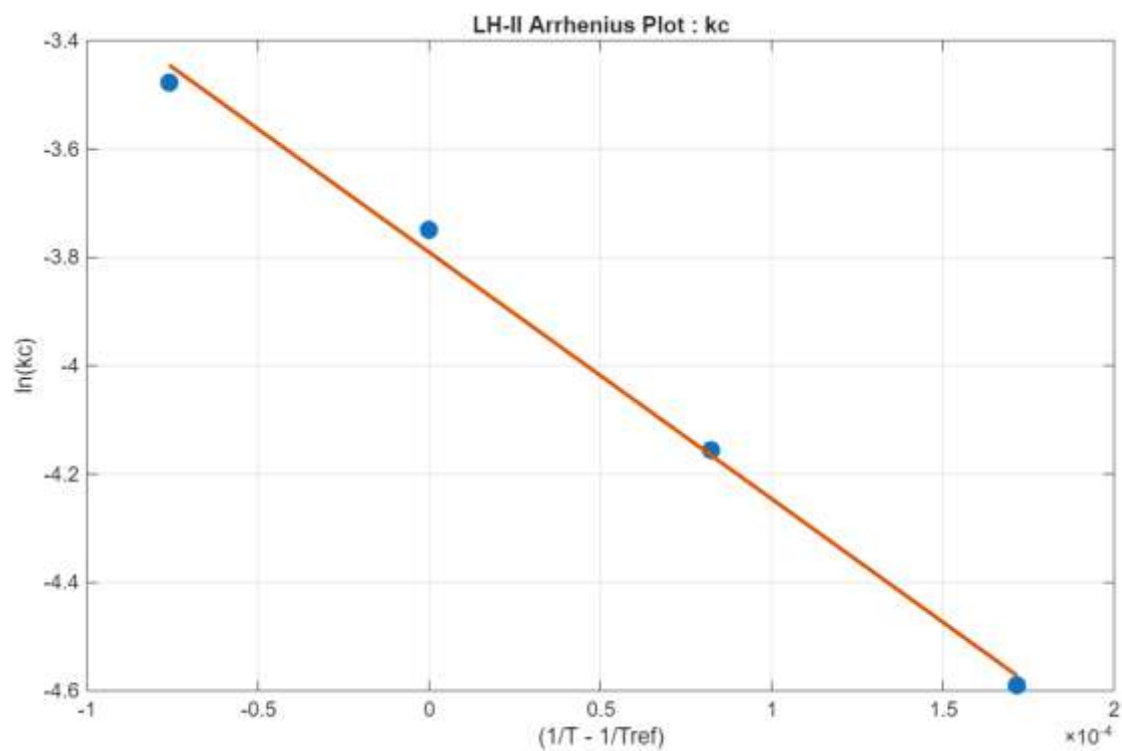


Figure 4.3 Arrhenius plot for furfural hydrogenation rate constant (k_c) using the dual-site Langmuir–Hinshelwood (LH-II) model.

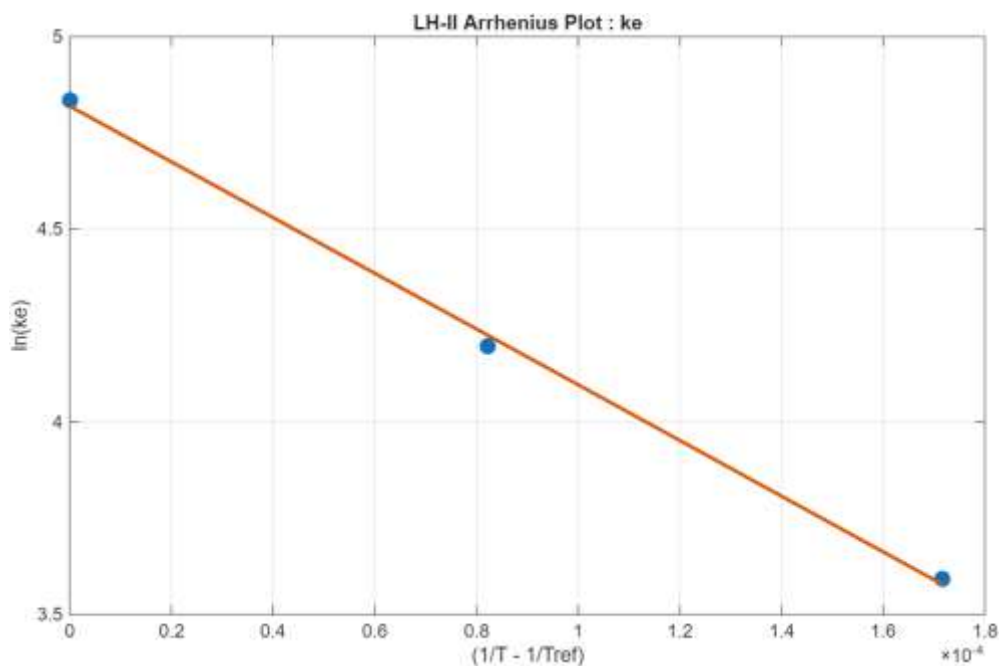


Figure 4.4 Arrhenius plot for deoxy hydrogenation rate constant (k_e) using the dual-site Langmuir–Hinshelwood (LH-II) model.

The parity plots comparing experimental and model-predicted reaction rates demonstrated good agreement for both Langmuir–Hinshelwood models. Most experimental data points were distributed closely around the 45° parity line, indicating satisfactory predictive capability of the developed kinetic models.

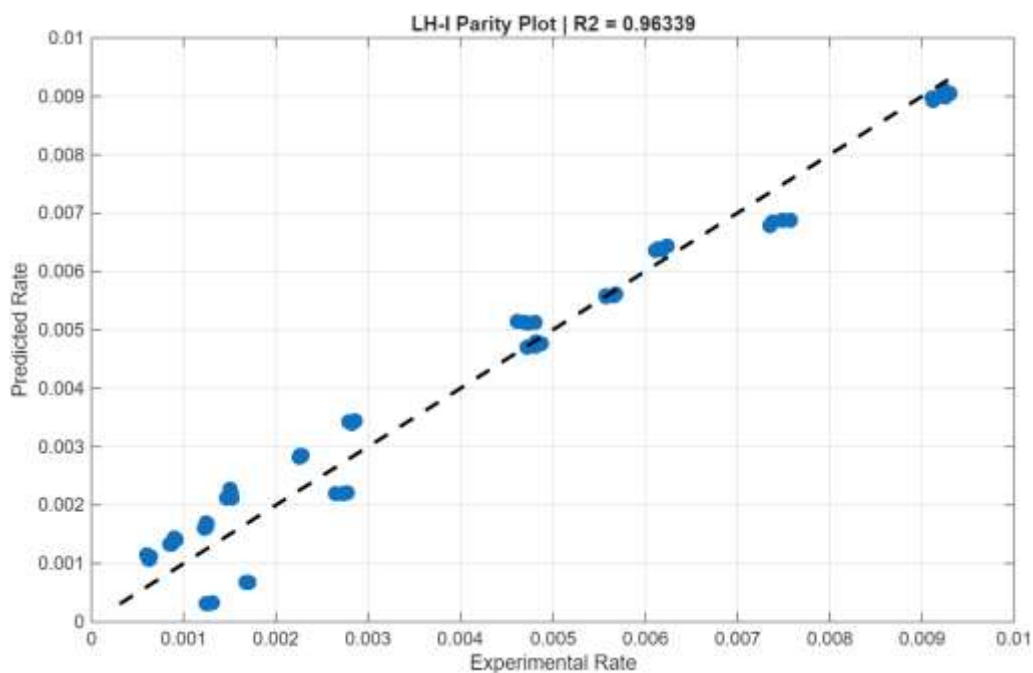


Figure 4.5 Parity plot comparing experimental and model-predicted reaction rates for the single-site Langmuir–Hinshelwood (LH-I) model.

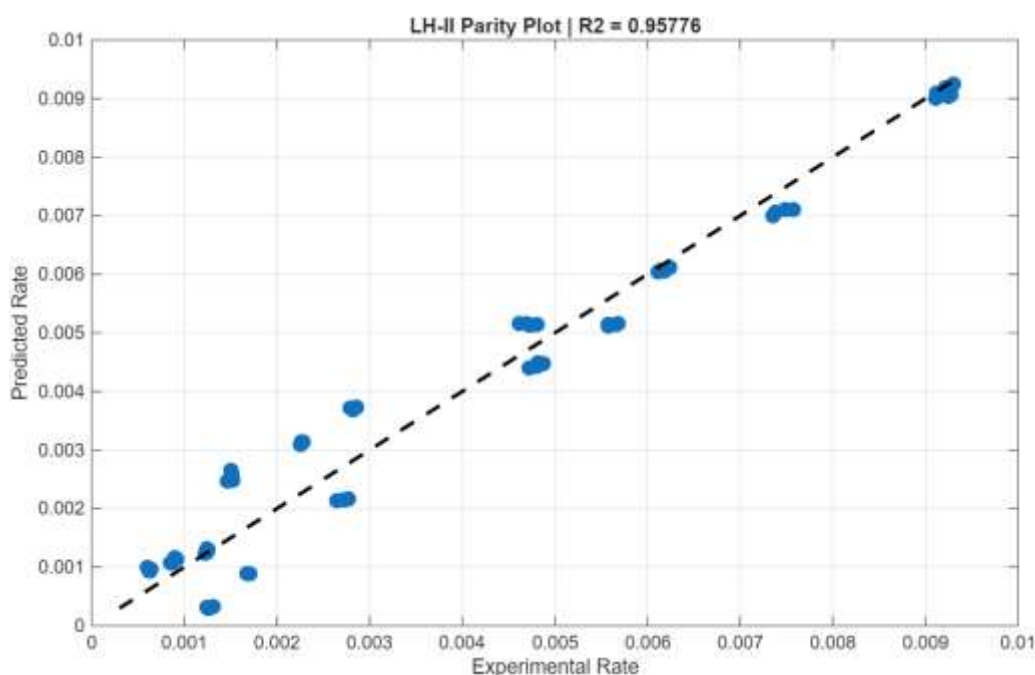


Figure 4.6 Parity plot comparing experimental and model-predicted reaction rates for the dual-site Langmuir–Hinshelwood (LH-II) model.

Overall, the dual-site Langmuir–Hinshelwood mechanism provided more physically meaningful adsorption parameters, lower parameter correlation, improved thermodynamic consistency, and better statistical significance compared to the single-site mechanism. Although both models adequately represented the experimental reactor data, the LH-II model demonstrated superior mechanistic reliability and better representation of catalyst surface interactions during furfural hydrogenation to 2-methylfuran.

5. CONCLUSION

In the present study, kinetic models for the production of 2-methylfuran from furfural were developed using power law and Langmuir–Hinshelwood mechanistic approaches. Single-site and dual-site Langmuir–Hinshelwood models were investigated to describe adsorption and surface reaction phenomena during furfural hydrogenation over CuFe_2O_4 and $\text{Cu}/\text{Fe}_2\text{O}_3$ catalysts. The estimated kinetic and thermodynamic parameters showed that the deoxy hydrogenation step required higher activation energy than furfural hydrogenation. Negative adsorption enthalpy and entropy values confirmed exothermic adsorption and reduced molecular randomness on the catalyst surface. Statistical analysis showed that the dual-site Langmuir–Hinshelwood model provided better thermodynamic consistency, lower parameter correlation, and improved parameter significance compared to the single-site model. Heat and mass transfer analyses confirmed the absence of transport limitations, indicating that the

observed reaction rates represented intrinsic catalytic kinetics. Overall, the dual-site Langmuir–Hinshelwood model provided the most reliable representation of the reaction mechanism and successfully described the catalytic conversion of furfural to 2-methylfuran.

NOMECLATURE

Symbol	Description	Units
C_A	Concentration of species A	mol/m ³
$C_{A,S}$	Concentration of species A at catalyst surface	mol/m ³
$C_{A,C}$	Concentration of species A at catalyst center	mol/m ³
C_s	Concentration of vacant active sites	mol/kg
C_t	Total active site concentration	mol/kg
C_{WP}	Weisz–Prater criterion	—
D_{eff}	Effective diffusivity	m ² /s
E_i	Activation energy	kJ/mol
F_i	Molar flow rate of species (i)	mol/h
F_T	Total molar flow rate	mol/h
h	Heat transfer coefficient	W/m ² K
ΔH_i	Heat of adsorption	kJ/mol
ΔH_R	Heat of reaction	kJ/mol
K_i	Adsorption equilibrium constant	l/bar
K_i	Reaction rate constant	Various
k_{ref}	Rate constant at reference temperature	Various
k_C	Mass transfer coefficient	m/s
P_i	Partial pressure of species (i)	bar
P_T	Total pressure	bar

Symbol	Description	Units
R	Universal gas constant	J/mol K
R_C	Catalyst particle radius	m
r_i	Reaction rate of species (i)	mol/h g_cat
-r_A,obs	Observed reaction rate	mol/kg s
R ²	Coefficient of determination	—
RMSE	Root mean square error	—
SSE	Sum of squared errors	—
T	Reaction temperature	K
T_ref	Reference temperature	K
W	Catalyst weight	g
WHSV	Weight hourly space velocity	1/h
X_FFR	Conversion of furfural	—
Y_MF	Yield of 2-methylfuran	—
Y_FAL	Yield of furfuryl alcohol	—
ΔS_i	Entropy change of adsorption	J/mol K
λ_{eff}	Effective thermal conductivity	W/m K
ρ_b	Bulk catalyst density	Kg/m ³

Species Abbreviations

Abbreviation	Species	Abbreviation	Species
FFR	Furfural	H	Dissociated Hydrogen
FAL	Furfuryl Alcohol	LH-I	Single-site Langmuir–Hinshelwood model
MF or 2-MF	2-Methylfuran	LH-II	Dual-site Langmuir–Hinshelwood model
H ₂ O	Water		

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